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COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.5 Introduction to Network Layer Protocols

- DHCP
- ICMP

4.6 IPv6 Addressing

4.7 Introduction to Routing Algorithms

That's actually **two** questions:

1. Q: How does a *host* get IP address within its network (host part of address)?
2. Q: How does a *network* get IP address for itself (network part of address)

How does *host* get IP address?

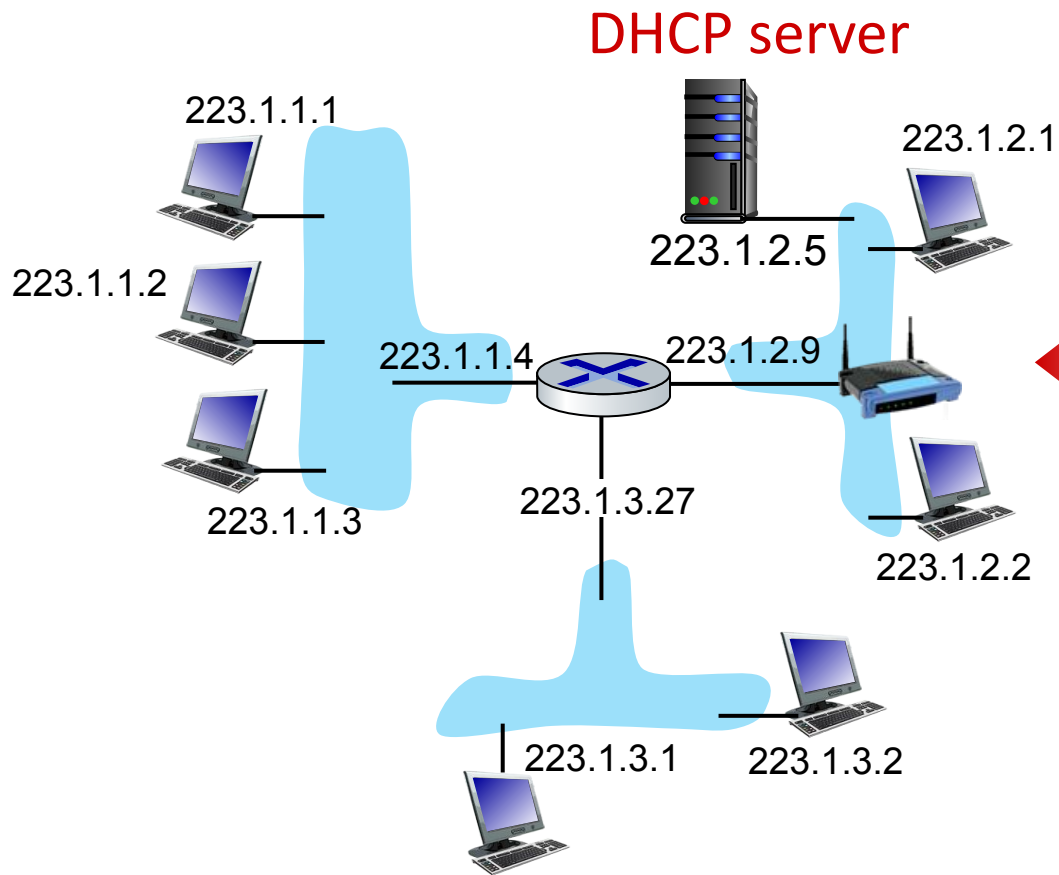
- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- **DHCP**: Dynamic Host Configuration Protocol: dynamically get address from as server
 - “plug-and-play”

goal: host *dynamically* obtains IP address from network server when it “joins” network

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

DHCP overview:

- host broadcasts **DHCP discover** msg [optional]
- DHCP server responds with **DHCP offer** msg [optional]
- host requests IP address: **DHCP request** msg
- DHCP server sends address: **DHCP ack** msg

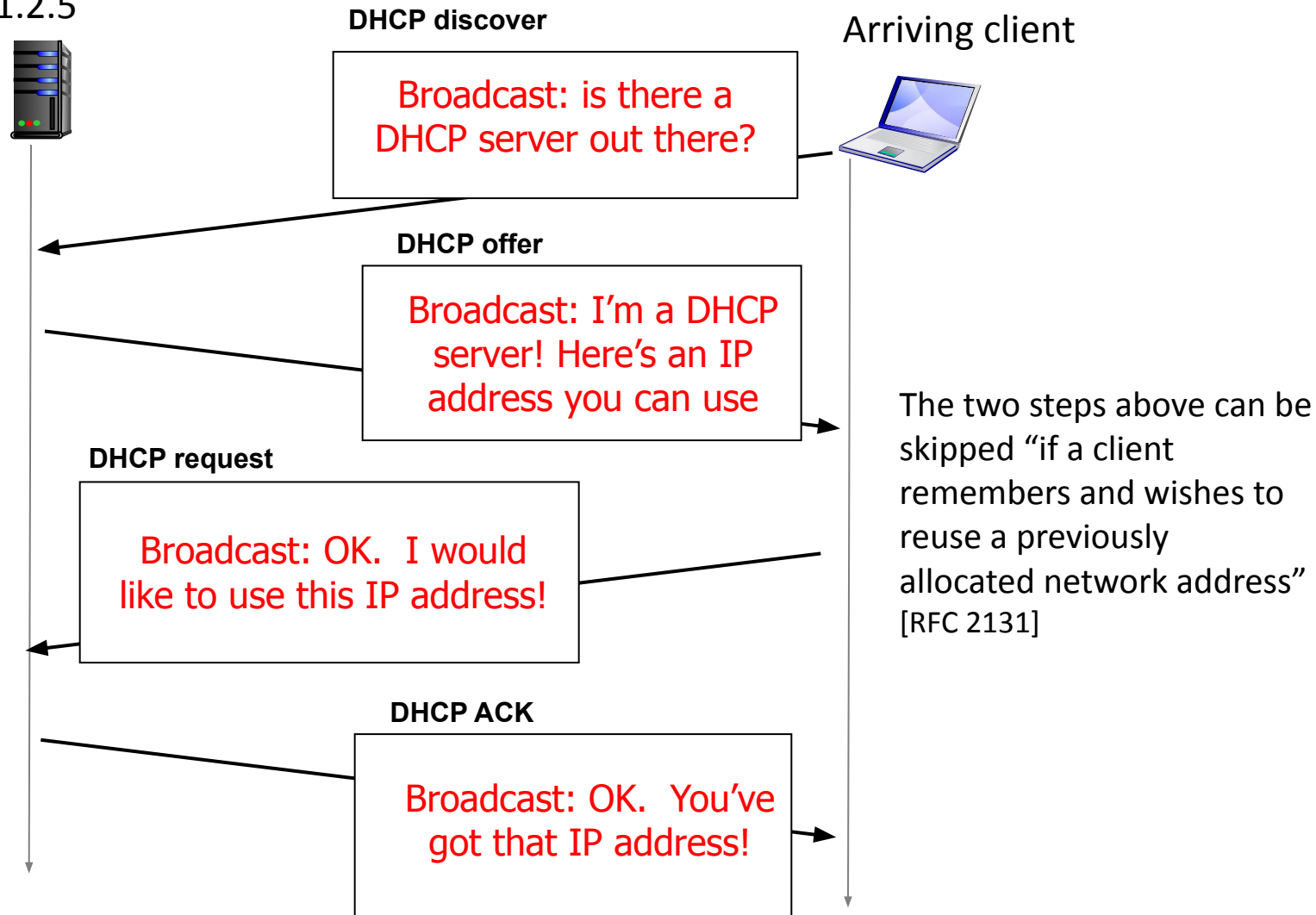


Typically, DHCP server will be co-located in router, serving all subnets to which router is attached



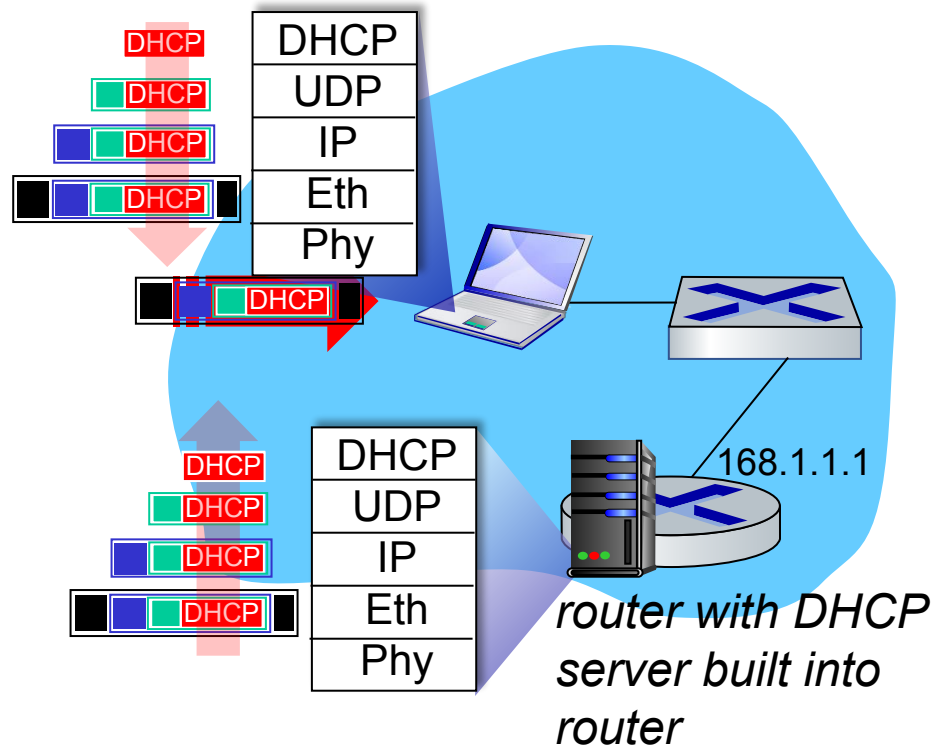
arriving **DHCP client** needs address in this network

DHCP server: 223.1.2.5

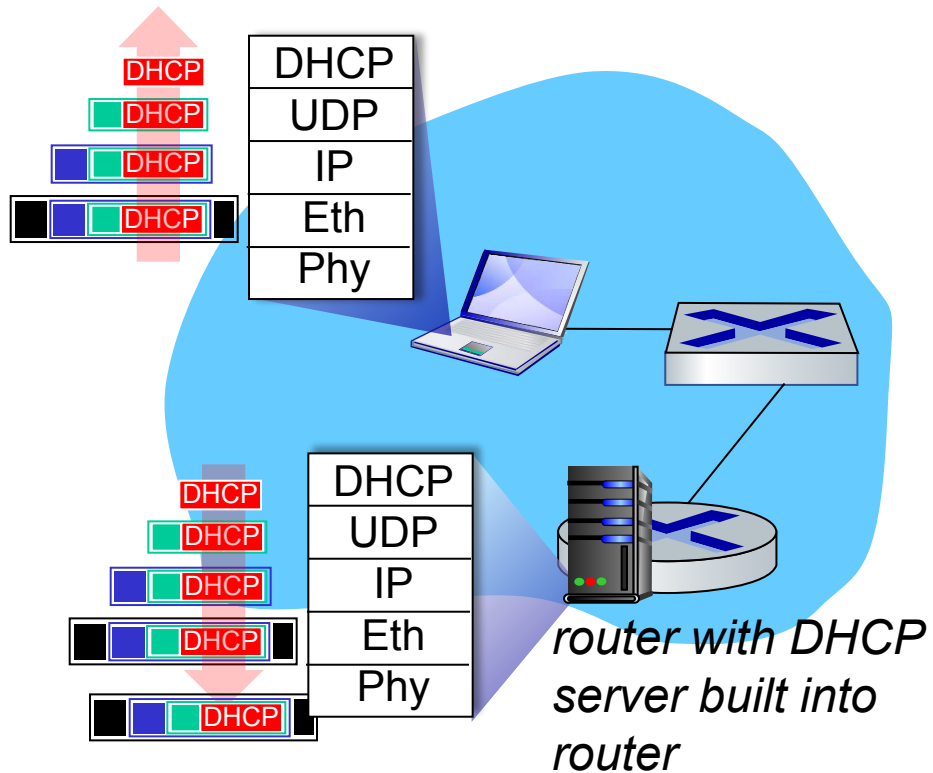


DHCP can return more than just allocated IP address on subnet:

- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)



- Connecting laptop will use DHCP to get IP address, address of first-hop router, address of DNS server.
- DHCP REQUEST message encapsulated in UDP, encapsulated in IP, encapsulated in Ethernet
- Ethernet frame broadcast (dest: FFFFFFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP



- DCP server formulates DHCP ACK containing client's IP address, IP address of first-hop router for client, name & IP address of DNS server
- encapsulated DHCP server reply forwarded to client, demuxing up to DHCP at client
- client now knows its IP address, name and IP address of DNS server, IP address of its first-hop router

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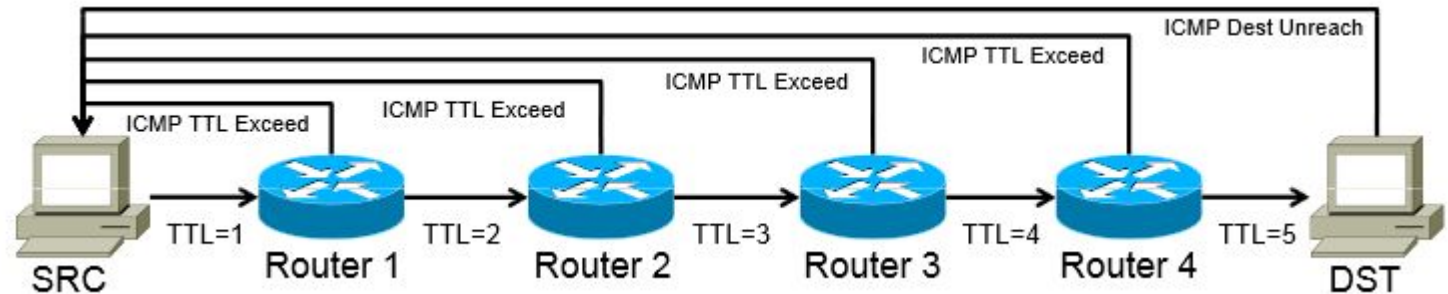
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- Used by hosts and routers to communicate network-layer information.
- Typical use of ICMP is for error reporting.
- Eg: “Destination network unreachable”
- ICMP messages – carried as IP payloads.
- ICMP is often considered part of IP, but architecturally it lies just above IP.
- ICMP messages have a type and a code field.

<u>Type</u>	<u>Code</u>	<u>description</u>
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

- Perform – Network diagnostics
- Utilities traceroute and ping – use ICMP.
- Other examples of ICMP messages:
 - Source quench message (congestion)
 - Parameter problem (bad IP header)
 - Time exceeded message (TTL)
 - Destination unreachable
 - Redirection message





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THANK YOU

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